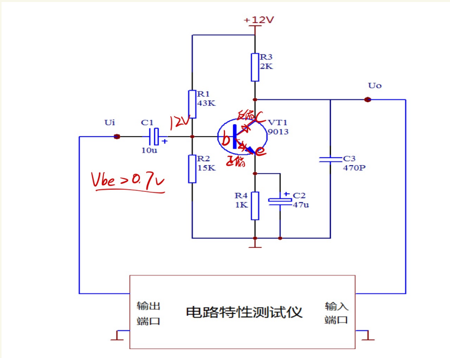



$R_i = R_1 // R_2 // [rbe + (1+\beta) Re]$

$= 43k // 15k // \cdots$

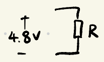
$= 11.12k // \cdots$

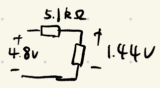
$\approx 11.12k$

$R_0 \approx R_3 \approx 2k\Omega$

Normal: $V_0 = 4.80V$.

5.08kΩ: $V_0 = 1.44V$.




5.1k : 4.8 - 1.44

$\frac{R_i}{1.44} = \frac{5.1k}{4.8-1.44}$

$R_i = 2.185.7 k\Omega$

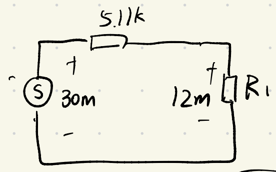
4.40V

1.44V

2.96

4.4V

1.44




$5.11 \times \frac{12}{18} = 3.4k\Omega$

$V_0 = 1.34V$

4.4V